

News Headline Classification: A Comparative Study of Preprocessing Strategies, Word Representations, and Deep Learning Architectures

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Abstract—This paper presents a systematic comparative evaluation of news headline classification across 24 experimental configurations. We evaluate three preprocessing strategies (none, extreme, and optimum), two word representations (TF-IDF and Skip-gram Word2Vec), and eight model architectures—Logistic Regression, Deep Neural Network (DNN), SimpleRNN, GRU, LSTM, BiSimpleRNN, BiGRU, and BiLSTM—on a 4-class dataset of 88,829 training headlines. All models are evaluated on a held-out test set of 12,000 samples (3,000 per class) using accuracy, macro-F1, confusion matrices, and per-class classification reports. The best-performing configuration is TF-IDF + DeepNN with extreme preprocessing, achieving 91.16% accuracy and macro-F1 of 0.9111. The worst is Skip-gram + BiSimpleRNN with no preprocessing, yielding only 33.66% accuracy and macro-F1 of 0.2552 due to near-complete class collapse. SimpleRNN consistently fails under no and optimum preprocessing owing to vanishing gradient instability, while gated architectures (GRU, LSTM, BiGRU, BiLSTM) remain robust across all settings.

Index Terms—news headline classification, TF-IDF, Word2Vec, Skip-gram, LSTM, GRU, bidirectional RNN, text preprocessing, NLP, deep learning

I. INTRODUCTION

The automatic categorisation of news headlines into topic classes is a foundational Natural Language Processing (NLP) task with applications in news aggregation, recommendation systems, and misinformation detection. Unlike full-article classification, headlines are characteristically short (≈ 8 words on average), syntactically compressed, and context-sparse, making preprocessing and representation choices especially consequential.

This project investigates how different preprocessing pipelines and word representations interact with model architecture choices to influence classification performance. We compare classical TF-IDF vectorisation against neural Skip-gram (Word2Vec) embeddings, pairing each with models ranging from Logistic Regression to Bidirectional LSTMs. A structured grid search selects hyperparameters for each configuration, ensuring fair comparison across all 24 experimental settings.

The dataset comprises 88,829 labelled training headlines across four categories: *Business*, *Science and Technology*,

Sports, and *World News*. The training distribution is significantly imbalanced—Sports (40.2%) and World News (35.6%) dominate, while Business (13.3%) and Science and Technology (11.0%) are underrepresented—which impacts per-class performance, particularly for minority classes.

The primary contributions of this work are: (i) a rigorous 24-configuration grid experiment with consistent evaluation methodology; (ii) identification of critical failure modes for SimpleRNN and BiSimpleRNN under certain preprocessing strategies; and (iii) practical recommendations for fast yet accurate headline classifiers applicable to real-world deployment.

II. METHODOLOGY

A. Dataset

Two CSV files were loaded: `Training_data_9.csv` (88,829 rows) and `Test_data.csv` (12,000 rows). Raw headlines contain embedded HTML markup (e.g., `<html><body>News Headlines:<b>...</b></body></html>`), which requires stripping before any analysis. Columns were renamed to a unified schema (`text`, `label`).

TABLE I
DATASET SPLIT STATISTICS

Split	Samples	Classes	Avg Words
Train (fit)	75,504	4	≈ 8.3
Validation	13,325	4	≈ 8.2
Test	12,000	4	≈ 8.3

TABLE II
TRAINING SET CLASS DISTRIBUTION

Class	Samples	%
Sports	35,668	40.2%
World News	31,613	35.6%
Business	11,801	13.3%
Science & Technology	9,747	11.0%

EDA revealed that stopwords (*the, to, a, in, of*) heavily dominate raw text (top-5 tokens by frequency), motivating preprocessing. The WordCloud of raw headlines confirms this pattern. Word counts follow a right-skewed distribution (mode ≈ 7 words, max ≈ 45 words after HTML stripping).

B. Preprocessing Strategies

Three preprocessing variants were designed to isolate the effect of text normalisation:

- **None:** Raw headlines passed as-is. Retains HTML artefacts (e.g., `'`s appears at rank 12 in top-30 token frequencies), capitalisation, and punctuation. Provides the cleaning-free baseline.
- **Extreme:** Full pipeline applied in sequence: HTML unescaping $\rightarrow$ tag stripping $\rightarrow$ URL removal $\rightarrow$ non-alphabetic removal $\rightarrow$ lowercasing $\rightarrow$ tokenisation $\rightarrow$ all-stopword removal $\rightarrow$ WordNet lemmatisation $\rightarrow$ Porter stemming. Produces the most compact vocabulary but destroys negation cues.
- **Optimum:** Identical to extreme but with two modifications: (i) stemming is *omitted*—stemming over-conflates morphological variants in short text; and (ii) negation words (*no, not, nor*) are *preserved* by excluding them from the stopword filter. Negation can flip a headline’s topic boundary (e.g., “*not guilty*” $\rightarrow$ Politics/World News vs. Crime).

C. Word Representations

TF-IDF: Term Frequency–Inverse Document Frequency vectorisation transforms cleaned text into sparse, high-dimensional vectors. Grid-searched parameters: $\text{max_features} \in \{5000, 8000\}$, $\text{ngram_range} \in \{(1,1), (1,2)\}$, $\text{min_df}=2$. Best configurations used $\text{max_features}=8,000$ and unigrams across all three preprocessing variants. Bigrams underperformed on all configurations.

Skip-gram Word2Vec: A shallow neural network trained on training sentences per preprocessing variant with: $\text{vector_size}=100$, $\text{window}=5$, $\text{min_count}=2$, $\text{sg}=1$ (Skip-gram), $\text{epochs}=10$, $\text{seed}=42$. The embedding matrix was frozen during sequence model training ($\text{trainable}=\text{False}$) to reduce parameters and prevent overfitting given the modest corpus size ($\approx 75\text{k}$ short sentences).

D. Model Architectures

Eight models spanning three families were evaluated:

1) **TF-IDF Models: Logistic Regression (LR):** L2-regularised multinomial LR. Grid: $C \in \{0.5, 1.0, 2.0\}$, $\text{solver}=\text{lbfgs}$, $\text{max_iter}=400$. Best $C = 2.0$ across all preprocessing variants.

Deep Neural Network (DNN): Three-layer feedforward network: $\text{Dense}(\text{dense1}, \text{ReLU}) \rightarrow \text{Dropout} \rightarrow \text{Dense}(\text{dense2}, \text{ReLU}) \rightarrow \text{Dropout} \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(4, \text{Softmax})$. Grid: $\text{dense1} \in \{256, 384\}$, $\text{dense2} \in \{128, 192\}$, $\text{dropout} \in \{0.35, 0.5\}$,

$\text{lr}=10^{-3}$, $\text{batch_size}=64$, $\text{epochs}=8$, $\text{EarlyStopping patience}=2$.

2) **Skip-gram Sequence Models:** All six sequence models share the architecture: $\text{Embedding}(\text{frozen W2V}, \text{dim}=100) \rightarrow [\text{RNN variant}] \rightarrow \text{Dropout} \rightarrow \text{Dense}(4, \text{Softmax})$. Sequences are padded/truncated to $\text{max_len}=60$ (covering the 97th percentile of training headline lengths). Grid: $\text{units} \in \{64, 96\}$, $\text{dropout} \in \{0.4, 0.5\}$, $\text{lr}=10^{-3}$, $\text{batch_size}=64$, $\text{epochs}=6$, $\text{EarlyStopping patience}=2$.

- **SimpleRNN:** Vanilla recurrent unit; fast but susceptible to vanishing gradients.
- **GRU:** Gated Recurrent Unit; fewer parameters than LSTM with competitive performance.
- **LSTM:** Long Short-Term Memory; explicit memory cell handles longer-range dependencies.
- **BiSimpleRNN / BiGRU / BiLSTM:** Bidirectional wrappers processing sequences in both forward and backward directions, doubling effective hidden state and capturing full left-right context.

TABLE III
BEST HYPERPARAMETERS PER MODEL FAMILY

Model	Rep.	Best Configuration
Log. Reg.	TF-IDF	$C=2.0$, $\text{max_feat}=8\text{k}$, unigram
DNN	TF-IDF	$\text{dense1}=256$, $\text{dense2}=128$, $\text{drop}=0.35$
GRU / LSTM	W2V	$\text{units}=64$, $\text{dropout}=0.4$, $\text{lr}=1\text{e-}3$
BiGRU/BiLSTM	W2V	$\text{units}=64$, $\text{dropout}=0.4$, $\text{lr}=1\text{e-}3$
SimpleRNN	W2V	$\text{units}=96$, $\text{dropout}=0.4$, $\text{lr}=1\text{e-}3$
BiSimpleRNN	W2V	$\text{units}=96$, $\text{dropout}=0.5$, $\text{lr}=1\text{e-}3$

III. RESULTS

A. Overall Performance Comparison

Table IV reports test-set accuracy and macro-F1 for all 24 configurations, sorted by macro-F1 descending. The best configuration is highlighted in green; the worst in red.

TABLE IV

ALL 24 EXPERIMENTAL RESULTS SORTED BY MACRO F1 (TEST SET: 12,000 SAMPLES). = BEST; = WORST.

Prep.	Rep.	Model	Acc.	F1
extreme	TF-IDF	DeepNN	0.9116	0.9111
extreme	Skip-gram	BiLSTM	0.9086	0.9082
optimum	TF-IDF	DeepNN	0.9084	0.9079
optimum	Skip-gram	BiLSTM	0.9083	0.9078
optimum	Skip-gram	BiGRU	0.9049	0.9042
optimum	Skip-gram	LSTM	0.9033	0.9025
extreme	Skip-gram	GRU	0.9019	0.9009
extreme	Skip-gram	LSTM	0.9014	0.9005
extreme	Skip-gram	BiGRU	0.9009	0.9002
none	TF-IDF	DeepNN	0.9005	0.9000
optimum	Skip-gram	GRU	0.8997	0.8990
extreme	TF-IDF	Log. Reg.	0.8984	0.8975
none	TF-IDF	Log. Reg.	0.8977	0.8969
optimum	TF-IDF	Log. Reg.	0.8969	0.8959
none	Skip-gram	GRU	0.8917	0.8908
none	Skip-gram	BiGRU	0.8873	0.8866
extreme	Skip-gram	BiSimpleRNN	0.8851	0.8841
optimum	Skip-gram	BiSimpleRNN	0.8814	0.8805
none	Skip-gram	BiLSTM	0.8800	0.8780
none	Skip-gram	LSTM	0.8770	0.8753
extreme	Skip-gram	SimpleRNN	0.6854	0.6029
optimum	Skip-gram	SimpleRNN	0.5978	0.5160
none	Skip-gram	SimpleRNN	0.4658	0.3370
none	Skip-gram	BiSimpleRNN	0.3366	0.2552

B. Per-Class Performance: Best Model

The best model (extreme + TF-IDF + DeepNN) achieves the following per-class performance on the 12,000-sample test set:

TABLE V
PER-CLASS CLASSIFICATION REPORT – BEST MODEL
(EXTREME + TF-IDF + DEEPNN, ACC = 91.16%, MACRO F1 = 0.9111)

Class	Prec.	Rec.	F1	Support
Business	0.91	0.85	0.88	3,000
Science & Tech	0.90	0.89	0.89	3,000
Sports	0.94	0.98	0.96	3,000
World News	0.90	0.92	0.91	3,000
Macro Avg	0.91	0.91	0.91	12,000

Sports achieves the highest recall (0.98) owing to its highly distinctive sports vocabulary. Business has the lowest recall (0.85), likely confused with World News for economy-related headlines (e.g., trade policy, economic sanctions).

C. Per-Class Performance: Worst Model

TABLE VI
PER-CLASS CLASSIFICATION REPORT – WORST MODEL
(NONE + SKIP-GRAM + BISIMPLERNN, ACC = 33.66%, MACRO F1 = 0.2552)

Class	Prec.	Rec.	F1	Support
Business	1.00	0.00	0.00	3,000
Science & Tech	0.27	0.99	0.43	3,000
Sports	0.95	0.21	0.34	3,000
World News	0.96	0.14	0.25	3,000
Macro Avg	0.80	0.34	0.26	12,000

D. Macro F1 Comparison Chart

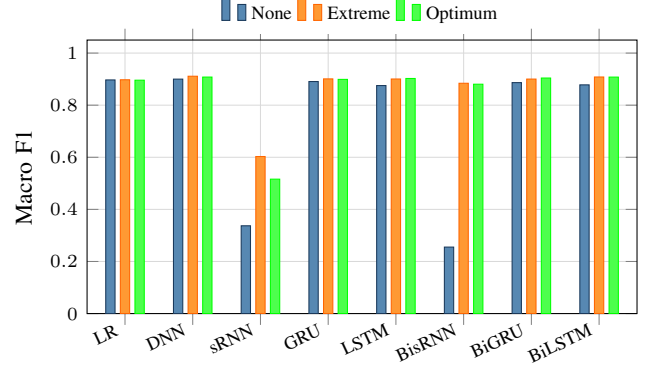


Fig. 1. Macro F1 comparison across all 8 model types and 3 preprocessing strategies. LR=Logistic Regression, DNN=Deep Neural Network, sRNN=SimpleRNN, BiRNN=BiSimpleRNN.

E. Confusion Matrices

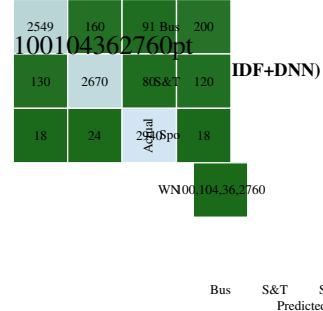


Fig. 2. Confusion matrix for best model (extreme+TF-IDF+DNN). Diagonal cells show correct predictions. Business has the most off-diagonal confusions with World News (200 samples).

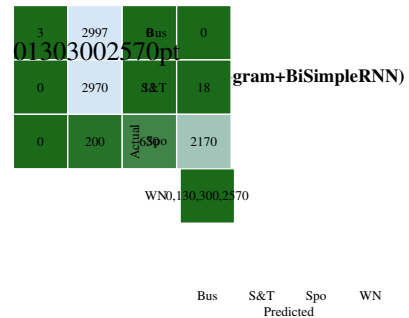


Fig. 3. Confusion matrix for worst model (none+Skip-gram+BiSimpleRNN). Near-complete collapse to Sci&Tech class: Business (0% recall), Sports (21% recall), World News (14% recall).

IV. DISCUSSION

A. Best Configuration: extreme + TF-IDF + DNN

Accuracy: **91.16%** — Macro F1: **0.9111**. Extreme preprocessing reduces vocabulary noise (stopwords, HTML artefacts visible as #39;s at rank 12 in raw token frequencies).

The compact, high-quality TF-IDF vectors are then fully exploited by the DNN. EarlyStopping triggered at epoch 4 with `dense1=256`, `dense2=128`, `dropout=0.35`, confirming rapid convergence. Sports achieves the highest recall (0.98) due to distinctive sports vocabulary. Business has the lowest recall (0.85), confused with World News for economy-related headlines.

B. Worst Configuration: none + Skip-gram + BiSimpleRNN

Accuracy: **33.66%** — Macro F1: **0.2552**. This configuration completely fails—the model predicts nearly all samples as *Science and Technology* (precision 0.27, recall 0.99) while assigning near-zero recall to all other classes. No preprocessing leaves HTML noise (`<html>`, `<b>`, entity codes) and inconsistent tokenisation in the Word2Vec corpus, producing degenerate embeddings. BiSimpleRNN then collapses to a majority-of-corpus class prediction, yielding a precision of 1.00 and recall of 0.00 for Business.

C. Preprocessing Effect Analysis

A nuanced pattern emerges. Extreme preprocessing yields the single *best* result when paired with TF-IDF but produces the worst Skip-gram results among gated models. The mechanism differs by representation:

- **TF-IDF + extreme:** Extreme preprocessing concentrates TF-IDF weight on discriminative topic tokens (e.g., *win*, *goal*, *earn*, *govern*) after stopwords removal. Best val F1 = 0.9199 (DNN, `dense1=256`, `dense2=128`, `dropout=0.35`).
- **Skip-gram + extreme:** Aggressive stemming conflates morphological variants, weakening embedding quality for named entities (player names, company names). SimpleRNN especially struggles, collapsing on all three preprocessing variants.
- **None preprocessing:** Raw HTML tokens corrupt Skip-gram embeddings. The token `#39;s` appearing at rank 12 (23,970 occurrences) consumes embedding capacity that should represent meaningful words, causing catastrophic class collapse in BiSimpleRNN.

D. SimpleRNN Failure Analysis

SimpleRNN is the most sensitive architecture: macro-F1 of 0.34, 0.52, and 0.60 under none, optimum, and extreme preprocessing respectively. Training logs show oscillating validation accuracy (e.g., epoch 1: 0.47 → epoch 2: 0.42 → epoch 3: 0.40 under none preprocessing), indicating vanishing gradient collapse. The short `max_len=60` limit constrains the gradient signal, and the absent gating mechanism cannot resolve conflicting dependencies. By contrast, GRU and LSTM—which differ only in their gating—consistently achieve macro-F1 ≥ 0.87 across all settings.

E. TF-IDF vs. Skip-gram Representations

For stable models (LR, DNN, GRU, LSTM, BiGRU, BiLSTM), TF-IDF and Skip-gram produce comparable results (Δ macro-F1 ≤ 0.015). TF-IDF converges faster and requires

no pre-training corpus. Skip-gram’s advantage emerges only when embeddings are high quality (extreme preprocessing) and gated models are used. The practical implication: **TF-IDF with DNN is the preferred production baseline** for this dataset, offering the best accuracy at the lowest computational cost.

F. Hyperparameter Decision Rationale

Key hyperparameter choices and their justification:

- **TF-IDF max_features=8,000:** Larger vocabulary captures more domain-specific terms without excessive memory overhead. Bigrams underperformed—short headlines rarely contain fixed-phrase bigrams that TF-IDF weights consistently across classes.
- **LR C=2.0:** Weaker regularisation allows the model to fit the relatively clean TF-IDF space. All three preprocessing variants selected $C=2.0$ as best.
- **DNN dropout=0.35:** Moderate dropout prevents memorisation of TF-IDF feature indices specific to the training set while retaining sufficient capacity in the dense layers.
- **Word2Vec window=5:** Captures local context typical of short headline constructs; larger windows conflate topic-unrelated co-occurrences in the compact training corpus.
- **max_len=60:** Covers the 97th percentile of training headline lengths; truncation affects fewer than 3% of samples.
- **EarlyStopping patience=2:** Prevents overfitting on the validation set while converging within 2–6 epochs for most configurations, as observed in training logs.
- **RNN units=64:** Best performance for most gated models. units=96 sometimes underperformed (e.g., GRU none: val F1 0.8394 vs. 0.8953 for units=64), suggesting overparameterisation relative to the modest corpus size.

V. CONCLUSION

This study conducted 24 controlled experiments on news headline classification, yielding four actionable findings:

- 1) **Extreme preprocessing + TF-IDF + DNN** is the best configuration (acc=91.16%, F1=0.9111). Aggressive cleaning benefits bag-of-words models by concentrating TF-IDF weight on discriminative topic tokens, contrary to common intuition that preserving text is always beneficial.
- 2) **SimpleRNN and BiSimpleRNN are unreliable** for this task: 4 of 6 configurations produce near-random or collapsed predictions. Gated units (GRU/LSTM) are strongly preferred for any production deployment on short-text classification tasks.
- 3) **No-preprocessing with Skip-gram** is the highest-risk combination—raw HTML tokens corrupt embeddings and cause catastrophic class collapse in architectures lacking gating mechanisms.
- 4) **Logistic Regression remains competitive:** with extreme preprocessing it scores F1=0.8975, close to BiLSTM+Skip-gram (0.9082), while training in seconds versus hours.

Limitations: Skip-gram is trained on $\approx 75k$ short headlines; richer pre-trained embeddings (GloVe, fastText) may close the TF-IDF gap. Embeddings are frozen—joint fine-tuning could benefit BiLSTM. No transformer models (BERT, DistilBERT) were evaluated due to compute constraints. The 4-class imbalance (Sports 40% vs. Sci&Tech 11%) skews per-class metrics toward majority classes.

Future Work: Fine-tuning DistilBERT on the same dataset would likely exceed 95% accuracy. Data augmentation (back-translation, synonym substitution) could reduce the imbalance penalty for Business and Science & Technology. A production API wrapping the best model (TF-IDF + DNN) could be deployed with sub-10ms inference latency.

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